Arista Access Node IND-ACC-16 Routing Protocol Flap Resolution SOP

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5777

Description

The occurrence of the Routing Protocol Flap (RPF) failure domain on the Access Node IND-ACC-16, located at the Indiranagar Exchange, is primarily attributed to the interaction between the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon arises due to the complex interplay between the routing protocols, which, when misconfigured or experiencing network instability, can lead to a cascade of events resulting in an excessive number of routing protocol flaps. The BGP route churn is triggered by changes in the BGP routing table, such as new routes being advertised or existing routes being withdrawn. This can be further exacerbated by the OSPF adjacency loss, which occurs when neighboring routers fail to establish or maintain an OSPF adjacency. Meanwhile, the ISIS LSP flooding is a result of the ISIS protocol's attempts to reconverge the network, leading to an increase in LSP (Label Switched Path) messages being flooded across the network. These events collectively contribute to the RPF failure domain, which is characterized by an increase in the number of routing protocol flaps, leading to the generation of the ALARM_5777 alarm code. This SOP aims to diagnose and resolve the root cause of this failure domain, ensuring that the Access Node IND-ACC-16 operates within normal parameters.

Pre-requisite Checks

- check: Verify that all redundant links are operational and functioning correctly.

- check: Dispatch remote hands to the Indiranagar Exchange to perform a site inspection and confirm that the node is accessible.

- check: Ensure that all necessary permissions and access credentials have been granted to perform the required diagnostic and resolution tasks.

Potential Root Causes

Cause	Probability
BGP route churn triggered by misconfigured or unstable BGP peerings	High
OSPF adjacency loss due to misconfigured or unstable OSPF neighbor relationships	Medium
ISIS LSP flooding resulting from misconfigured or unstable ISIS protocol settings	Medium

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP neighbor table: ..., Description: This command is used to gather information about the BGP neighbor relationships and determine if any of the peers are experiencing issues that may be contributing to the BGP route churn.
```

```
Command: show ospf neighbor, Expected Output: OSPF neighbor table: ..., Description: This command is used to gather information about the OSPF neighbor relationships and determine if any of the neighbors are experiencing issues that may be contributing to the OSPF adjacency loss.

Command: show isis database, Expected Output: ISIS database: ..., Description: This command is used to gather information about the ISIS LSP database and determine if any of the LSPs are experiencing issues that may be contributing to the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify and correct BGP peerings, Command: show bgp summary | grep | awk '{print $4}', Description: This command is used to verify the BGP peerings and determine if any of the peers are experiencing issues. If a peer is experiencing issues, use the output from this command to correct the peerings.

Step 2: Step: Verify and correct OSPF neighbor relationships, Command: show ospf neighbor | grep | awk '{print $4}', Description: This command is used to verify the OSPF neighbor relationships and determine if any of the neighbors are experiencing issues. If a neighbor is experiencing issues, use the output from this command to correct the neighbor relationships.

Step 3: Step: Verify and correct ISIS protocol settings, Command: show isis database | grep | awk '{print $4}', Description: This command is used to verify the ISIS LSP database and determine if any of the LSPs are experiencing issues. If an LSP is experiencing issues, use the output from this command to correct the ISIS protocol settings.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP neighbor table: ..., Description: This command is used to verify that the BGP neighbor relationships have been corrected and are functioning correctly.

Command: show ospf neighbor, Expected Output: OSPF neighbor table: ..., Description: This command is used to verify that the OSPF neighbor relationships have been corrected and are functioning correctly.

Command: show isis database, Expected Output: ISIS database: ..., Description: This command is used to verify that the ISIS LSP database has been corrected and is functioning correctly.
```

Roll-back Procedures

- Step: Restore original BGP peerings, Command: copy running-config startup-config, Description: This command is used to restore the original BGP peerings and revert any changes made during the resolution process.

- Step: Restore original OSPF neighbor relationships, Command: copy running-config startup-config, Description: This command is used to restore the original OSPF neighbor relationships and revert any changes made during the resolution process.

- Step: Restore original ISIS protocol settings, Command: copy running-config startup-config, Description: This command is used to restore the original ISIS protocol settings and revert any changes made during the resolution process.